

Side-Scan Sonar as a Tool For Measuring Fish Populations: Current State of the Science and Future Directions

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Satellite image with side-scan sonar mosaics. Image credit: U.S. Geological Survey, State of Missouri, STL Imagery Consortium, Maxar, Microsoft.

Side-scan sonar (SSS) is a powerful tool that can be used to address many key questions in fisheries science. In principle, SSS uses dual transducers to transmit a narrow-beam, wide-angle acoustic signal as the survey vessel transits an area. The intensity of reflected sound is recorded to generate an image mosaic comprised of benthic substrates and targets in the water column, including organisms such as fish. Although SSS has been around for decades, recent advancements have opened new opportunities to leverage this technology to directly measure fish populations. In this paper, we review the current state of the science and identify opportunities to further refine SSS for fisheries applications.

EVOLUTION OF SIDE-SCAN SONAR AS A TOOL FOR FISHERIES SCIENCE

The first operable side-scan sonar (SSS) was reportedly constructed by the National Institute of Oceanography in England (Tucker and Stubbs 1961). Early systems were analog and relatively low frequency (e.g., 5–100 kHz), but higher frequency sonars (e.g., hundreds of kHz) were subsequently developed, providing greater resolution and digital recording. In the 1970s, SSS became commercially available and was primarily used for mapping geological features on the seafloor (Flemming 1976), but it was also successfully used to detect fish in open-ocean environments (Trevorrow and Claytor 1998). Sonar technology eventually integrated global navigation satellite systems during the 1980s. The ability of SSS to map substrate types and to resolve sonar targets both on the bottom and in the water column advanced as even higher frequency sonars (e.g., 400–1,600 kHz) and improved digital recording systems with post-processing software became available. In 2005, the first recreational-grade SSS products (i.e., SSS fishfinders) were made commercially available for recreational anglers and boaters. Advancements in recreational-grade SSS in recent years have enabled researchers to reliably quantify fish habitat features (Kaesler and Litts 2008, 2010; Kaesler et al. 2013) in shallow habitats (generally nearshore and inland waters less than 20 m) and have enabled biological targets to be identified and counted, with a level of resolution sufficient for surveying many fishes of management concern. The continuous progression in technology, user interface improvements, and reduced equipment costs (particularly for recreational-grade SSS) have provided substantive opportunities for wide-scale use in aquatic research and management.

Technological advances have created numerous opportunities to apply SSS in support of fisheries science, including in contexts where traditional approaches have failed to meet management needs. Population size structure and abundance data are critical to managing fish stocks. For some target populations and dynamic habitats, obtaining these data via physical capture methods is difficult, whereas SSS can be a more robust approach. Large swaths of water can be sampled with relatively low effort and minimal impact to fish, which are important considerations for surveying rare, threatened,

and endangered species. Early efforts focused on using SSS to count or characterize large schools of fish (Trevorrow and Claytor 1998), whereas more recent work has used the technology to directly enumerate individual fish for a growing array of taxa, with potential application to many management or conservation contexts. Side-scan sonar has been used to successfully quantify sturgeons *Acipenser* spp. (Flowers and Hightower 2013, 2015; Hughes et al. 2018), Alligator Gar *Atractosteus spatula* (Fleming et al. 2018), bigheaded carps *Hypophthalmichthys* spp. (Lawson et al. 2020), Paddlefish *Polyodon spathula* (Wolfenkoehler et al. 2023), and Smalltooth Sawfish *Pristis pectinata* (Papastamatiou et al. 2020) as well as other aquatic fauna, such as the Antillean manatee *Trichechus manatus manatus* (Guzman and Condit 2017). While large animals with distinctive profiles are most readily identified in SSS imagery, recent work with Shortnose Sturgeon *Acipenser brevirostrum* (Andrews et al. 2020) and bigheaded carps (Lawson et al. 2020; Rivera et al. 2023) demonstrated that the techniques can be extended to smaller organisms (e.g., 0.4–1.0 m long). In general, the examples provided indicate that SSS appears to perform best when imaging targets in relatively shallow water (primarily nearshore and inland habitats) underlain by fine-grained, homogeneous sediments.

Side-scan sonar has been used since 2012 to directly enumerate the number of large Gulf Sturgeon *A. oxyrinchus desotoi* within an index reach of the Apalachicola River system, Florida (Dula et al. 2022). Other studies have used repeated sampling to fit *N*-mixture models and estimate the abundance of sturgeon within specific reaches (Table 1; Hughes et al. 2018; Vine et al. 2019). However, spatially constrained counts have also been extended to reachwide or riverwide counts using a variety of statistical approaches. For example, SSS counts can be integrated with acoustic telemetry data to extend local counts to a broader population of interest. Andrews et al. (2020) related the total number of acoustic transmitters detected within the focal survey area to the total number of active transmitters in the system in order to extrapolate local sturgeon counts to an estuarywide abundance. In another example, Kazyak et al. (2020) combined sonar counts and acoustic telemetry data in a Bayesian hierarchical model to estimate the size of a spawning run of Atlantic Sturgeon

Table 1. Modeling frameworks that have been applied to analyze fish counts derived from side-scan sonar.

Modeling framework	Uses	Assumptions	Foundational citations	Applications of models
<i>N</i> -mixture	Repeated count data	Population closure. However, open frameworks exist	Royle (2004); Dail and Madsen (2011)	Flowers and Hightower (2015); Vine et al. (2019)
Distance	Allows for detection probability to vary with distance	Uniform underlying distribution; spatial variation due to observation process	Buckland et al. (2001)	Flowers and Hightower (2015); Wolfenkoehler et al. (2023)
Occupancy	Presence/absence only; target group absence	Detection probability is consistent across space and/or time	Mateo et al. (2010); MacKenzie et al. (2017)	Flowers and Hightower (2013)
Integrated models	Maximum flexibility; can incorporate disjoint data types (e.g., telemetry)	Variable; context and data dependent	Kéry and Royle (2020)	Kazyak et al. (2020)

A. oxyrinchus oxyrinchus. Those authors coupled sonar counts with telemetry detections in a focal reach to estimate the proportion of the spawning population that was fitted with an acoustic transmitter. They then inferred the total run size by extrapolating the count of tagged fish that were detected on a riverwide array using the estimate of the proportion of the population that was tagged. Even for traditional capture-based population monitoring programs, real-time to near real-time application of SSS can identify habitats and aggregations of target species, improve capture efficacy, and assist in avoiding hazardous underwater structure (e.g., wood debris) during boat navigation and gear deployment.

A REVIEW OF CURRENT APPROACHES AND CONSIDERATIONS

Although SSS has become increasingly accessible and relevant to fisheries science, there are several outstanding obstacles that remain as barriers to more widespread application. These obstacles primarily relate to the challenges and costs associated with the collection and analysis of sonar data. In the following sections, we will review current approaches and identify potential strategies to resolve these issues. We will work through the process in chronological order from initial study design to data collection and analysis.

Study Design

The objectives of a research project should be reflected in its study design, ensuring that there is a clear and deliberate path from data collection to analysis and interpretation. As SSS matures as a tool to quantify fish, greater consideration of study design may also be warranted (Radinger et al. 2019). Key considerations include movement ecology and behavior of the target species; the timing, extent, and frequency of surveys; the importance of key parameters, such as detection probability and target composition; and selection of an appropriate analytical approach.

A variety of study designs has been used to count fish with SSS. In general, most surveys have been implemented at a time and place where the target species is expected to be concentrated. Although some surveys aim to capture a snapshot in time (Andrews et al. 2020), many studies have used repeated surveys across time to increase statistical rigor, which also can allow for dynamics in local abundance to be estimated (Kazyak et al. 2020). The latter approach may be especially useful when target densities are low or when detection probability needs to be parameterized. It is beyond the scope of this paper to review all possible designs, many of which are analogous to traditional survey approaches (e.g., trawl surveys). Uncertainty in detection probability is not unique to sonar but is an important consideration for all fisheries sampling techniques. We note that fish detection probability and composition in sonar imagery are two key parameters to consider at the design stage. However, sonar research to reduce uncertainty is ongoing as the technology progresses and is applied more broadly.

A key challenge in using SSS to count fish is the uncertainty in image interpretation. The size and reflectivity of targets as well as the bottom habitat may impact the ability to accurately image and identify fish in sonar imagery. Detection probability of the target species may be high in some systems (Kazyak et al. 2020; Lawson et al. 2020), but in other applications it may be a critical factor to consider when estimating abundance. Imperfect detection probability may be assessed

through variability among repeated surveys, by using known targets, and/or via distance modeling to assess the uncertainty in gear detection. The uncertainty surrounding reader detection may be assessed across multiple independent reviewers or via the performance metrics of algorithm/artificial intelligence (AI)-based counts. If estimation of detection probability is not feasible or warranted, consistent methodology will help to ensure that results are comparable over time and space. In addition, some environments may contain taxa that are similar (e.g., in body size, body shape, and habitat selection) to the species of interest but that are not the focus of the study. In this scenario, supplemental data collection (e.g., the use of capture gears to estimate species and size composition) may be necessary to support later analysis and interpretation. We note that there are models that have been specifically developed to address false positives (Royle and Link 2006; Ferguson et al. 2015), but we are not aware of any published applications to sonar studies.

To be both effective and efficient, surveys need to have sufficient coverage across space and time to provide statistical rigor. We are unaware of any investigations to date that have formally assessed how much effort or sonar survey area is necessary to address research objectives. Evaluations are needed to ensure that the data have sufficient statistical power while minimizing the excessive use of resources, especially for applications that seek to enumerate fish over large spatial extents where exhaustive surveys are intractable. One approach to address this need would be to collect empirical data from pilot field studies and use those data to conduct power analyses. Standardization has also been promoted as an opportunity to increase the utility of fisheries surveys and ensure interoperability among data sets (Bonar et al. 2009). At present, there are no clear guidelines for standardizing sonar surveys, and the methods and equipment vary widely among research teams. At a minimum, we suggest that maintaining clear metadata relating to surveys (e.g., sonar type and settings [such as frequency and range], vessel speed, and post-processing methods) is important.

Equipment Selection

There are numerous commercially available SSS systems that can be broadly categorized into two separate classes. Survey-grade equipment typically is more robust, integrated with powerful data collection hardware and software, and capable of the broadest scan widths, but it typically requires a significant capital investment. In contrast, recreational-grade SSS equipment is marketed at a more accessible price point and interface (Figure 1; Table 2; e.g., Halmai et al. 2020). Although not mutually exclusive, survey-grade systems typically are deployed in a towfish configuration, which can help to modulate the influence of surface waves on the imagery but also limits detections to targets below the towfish depth (Figure 1). In contrast, recreational-grade sonar units typically employ a hull-mounted transducer, which is simpler to use and can be deployed in much shallower habitats (Figure 1; Kaeser et al. 2013; Lawson et al. 2020). Critically, both types of sonar systems can be effective for imaging individual fish targets in shallow habitats (e.g., generally less than 20 m). Regardless of the class of equipment used (Table 2), higher frequencies (e.g., 600–1,600 kHz) are generally required for imaging individual fish targets (e.g., Kazyak et al. 2020; Lawson et al. 2020). For studies focusing on smaller targets, the use of high frequency is especially critical to obtain adequate imagery for fisheries applications. However, there are

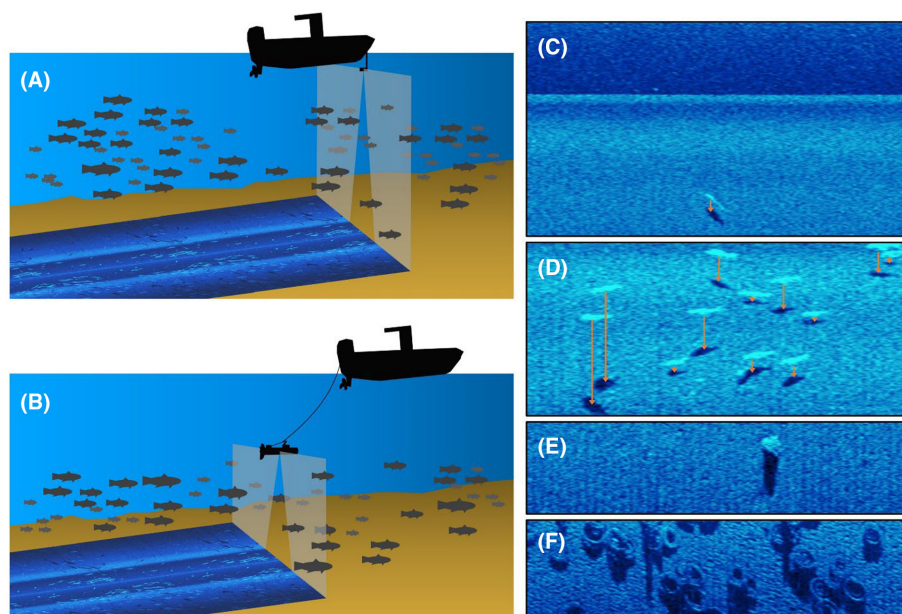


Figure 1. Illustrations of side-scan sonar deployments, including (A) a bow-mounted transducer (e.g., recreational-grade sonar) and (B) a towfish transducer tethered to the stern (e.g., survey-grade sonar). Imagery is truncated at a specific distance to account for signal attenuation. Additionally, a collection of imagery examples (i.e., recreational-grade sonar) depicting (C) the nadir zone transition to the substrate; (C), (D) fish targets off the substrate, with associated acoustic shadows indicated by the arrows; (E) a stump associated with a long acoustic shadow (contact with the acoustic return suggests contact with the substrate); and (F) acoustic returns from sunken tires on the substrate. Note that sonar beam angles and ensonified coverage can vary among manufacturers.

Table 2. A comparison of the characteristics of recreational- and survey-grade side-scan sonar units (with compressed high-intensity radar pulse [CHIRP]) for measuring fish populations. The recreational-grade sonar systems listed can simultaneously record other sonar channels at equal frequency, such as down-imaging, and are models commonly used by natural resource agencies and anglers in the United States. The EdgeTech 4125 survey-grade sonar units, which are most commonly used for fish enumeration, are currently available to 1,600 kHz and can simultaneously image at 600 kHz (Kazyak et al. 2020).

Characteristic	Sonar type	
	Recreational grade	Survey grade
Resolution	High-resolution options (up to 1,200 kHz)	Highest resolution options (up to 1,600 kHz)
Availability	Installed on many agency-owned and private vessels for navigation and “fish-finding”	Primary use for research surveys
Deployment	Typically hull-mounted and more suitable for shallower water	Typically deployed as a towfish, but can be mounted to the vessel
Ease of use	More portable and user-friendly interface	Bulkier equipment and cables. The interface requires greater training and technical understanding.
Software	Manufacturer-supported software for quantifying fish targets is currently limited. Third-party software options are available.	Manufacturer supported software and third party options available for quantifying fish targets.
Example equipment models	Humminbird (G4N: 1,200 kHz; Johnson Outdoors, Racine, Wisconsin), Garmin (ECHOMAP: 1,200 kHz; Olathe, Kansas), Lowrance (HDS Pro: 1,075 kHz, Tulsa, Oklahoma)	EdgeTech (4125P: 600/1,600 kHz; West Wareham, Massachusetts), Klein Marine Systems (5900: 600 kHz; Salem, New Hampshire), JW Fishers (450/900: 1,200 kHz; East Taunton, Massachusetts)

trade-offs to consider. High frequency has less effective range, particularly in turbid or turbulent conditions. We recommend pilot testing of the available options in the study system and, if possible, with the target species present to ascertain the optimal frequency and settings. Selected settings should be held constant across the study period to avoid introducing confounding variables into the analysis.

Some consideration of the survey vessel is also warranted. Beyond crew comfort and safety, more stable vessels will help to generate higher quality sonar imagery, especially if hull-mounted transducers are used. For survey-grade systems, it

is helpful to have a sheltered area onboard where the laptop and topside processing units can be safely set up. In addition, noises produced by the vessel may impact the behavior of some species, especially those that are known to be reactive to vessel traffic (e.g., bigheaded carps; Vetter et al. 2017). In such instances, small, quiet watercrafts may be an option to consider (Figure 2). Electric motors or trolling motors may be used to limit noise disturbance. For example, GPS-enabled trolling motors have recently become available on the recreational-grade market and may be a low-cost approach to quality control for tracking and velocity as well as to power

uncrewed kayaks for conducting SSS surveys (Figure 2). If resources are available, multiple semiautonomous kayaks may be released simultaneously to generate greater swath coverage per unit of time.

Data Collection

The timing and location of sonar surveys warrant careful consideration. Sonar surveys are most likely to be successful in environments where fish targets are more easily distinguished from other echo returns in the sonar imagery. In general, scanning locations where the focal populations occur over bottom substrates such as mud or sand are preferred. Rocky and/or highly variable substrates, woody debris, aquatic vegetation, and excessive sediment in the water column can make it difficult to identify targets in sonar imagery. Survey design and additional approaches that minimize movement responses should be employed to maximize image quality and reduce count biases—either positive (double counting) or negative (fish vessel avoidance). Such factors that could be important include the timing of surveys (day versus night; season), specifying the distances of reliable target acquisition from the survey vessel, and employing survey vessel types that can minimize disturbance behavior (e.g., Figure 2). Fish composition in open systems is known to have a strong association with season. For example, in paired SSS and gill net surveys of overwintering Shortnose Sturgeon on the Hudson River, researchers observed almost exclusively Shortnose Sturgeon throughout most of the season, but Striped Bass *Morone saxatilis* and Atlantic Sturgeon were present in late winter or early spring (Amanda Higgs, New York State Department of Environmental Conservation, personal communication). Coldwater periods may limit fish swimming during sonar surveys. Active swimming may lead to distorted or elongated echo returns from fish. Bigheaded carps are known to exhibit strong behavioral responses to vessels during the day (Figure 3). However, there is some evidence to suggest that boat avoidance responses may decrease at night (Ridgway



Figure 2. Recreational-grade side-scan sonar coupled with semiautonomous watercrafts (i.e., kayaks with GPS-enabled trolling motors) is being tested for use in surveying bigheaded carps. The side-scan sonar transducer is deployed off the stern. Note that the trolling motor must be compatible with the sonar head unit. Pre-programmed line coordinates are uploaded to the head unit to enable self-driving along survey transects at a constant velocity. Photo credit: U.S. Geological Survey.

et al. 2020), which is consistent with sonar studies demonstrating an increase in bigheaded carp abundance during the night compared to day (Ye et al. 2013).

Finally, it is important to consider how a given combination of sonar equipment and deployment strategy will impact the extent of the water column that is ensonified. Targets that are outside of the ensonified area (i.e., either above the transducer, outside of the beam angle, or beyond the effective range) cannot be imaged and will be unavailable for detection and enumeration.

Data Processing and Target Identification

A typical data processing flow is as follows: (1) recording of raw data; (2) initial processing, including bottom tracking, slant range corrections, and applying gain; (3) data conversion to spatially referenced images; (4) image pre-processing steps (e.g., image augmentation and filtering); and (5) fish detection (e.g., fish counting, sizing, and orientation). Raw sonar data are processed and converted into spatially referenced images either using the manufacturer's supporting software or third-party software options (Table 2). Several third-party software options are available for either survey- or recreational-grade systems; examples include SonarWiz (Chesapeake Technology, Los Altos, California), Survey Engine Sidescan+ (Coda Octopus Products, Orlando, Florida), SonarTRX Pro (Leraand Engineering, Honolulu, Hawaii), and PING-Mapper (Bodine et al. 2022). These range from free, open-source software to much more expensive alternatives, with varying capabilities. Open-source image software tools (e.g., ImageJ; Ferreira and Rasband 2019) can be applied in any number of ways to augment images (e.g., smooth, sharpen, or despeckle) and improve the contrast of fish echo returns relative to background noise. The occurrence of high-density substrates, nontarget materials, noisy ambience, or entrained air bubbles can obscure the echo returns from fish. Image analysis tools may be applied to batch filter or crop poor-quality images having high reflectance intensity as part of quality assurance and quality control procedures.

A major challenge is to identify targets within the imagery. In our experience, this is currently one of the most limiting factors to broader applications of SSS in fisheries. There are currently no standardized approaches, and research teams must identify a tractable approach that is consistent with available resources, expertise, and research objectives. Once the imagery has been prepared, target identification can be completed with either manual (i.e., human) reviewers or through automated software approaches. Manual analysis is a common approach. Reviewers may spend a great deal of time closely scrutinizing sonar imagery—a task that can be monotonous, particularly for large data sets or highly abundant fishes—and this approach presents challenges for maintaining consistency in target identification. Consistency among reviewers can be improved by using clearly defined criteria that are established prior to target identification, conducting training sessions that use sample imagery, and using high-quality computer monitors and ergonomic seating. Advancements in AI (e.g., machine learning and/or deep learning algorithms) offer a promising avenue to automate processing steps, reduce time constraints, and apply learned extraction criteria to sonar imagery (Tarling et al. 2022; Wei et al. 2022). Conceptual benefits include resources to limit human time, repeatable science, and near real-time counts and reporting from survey

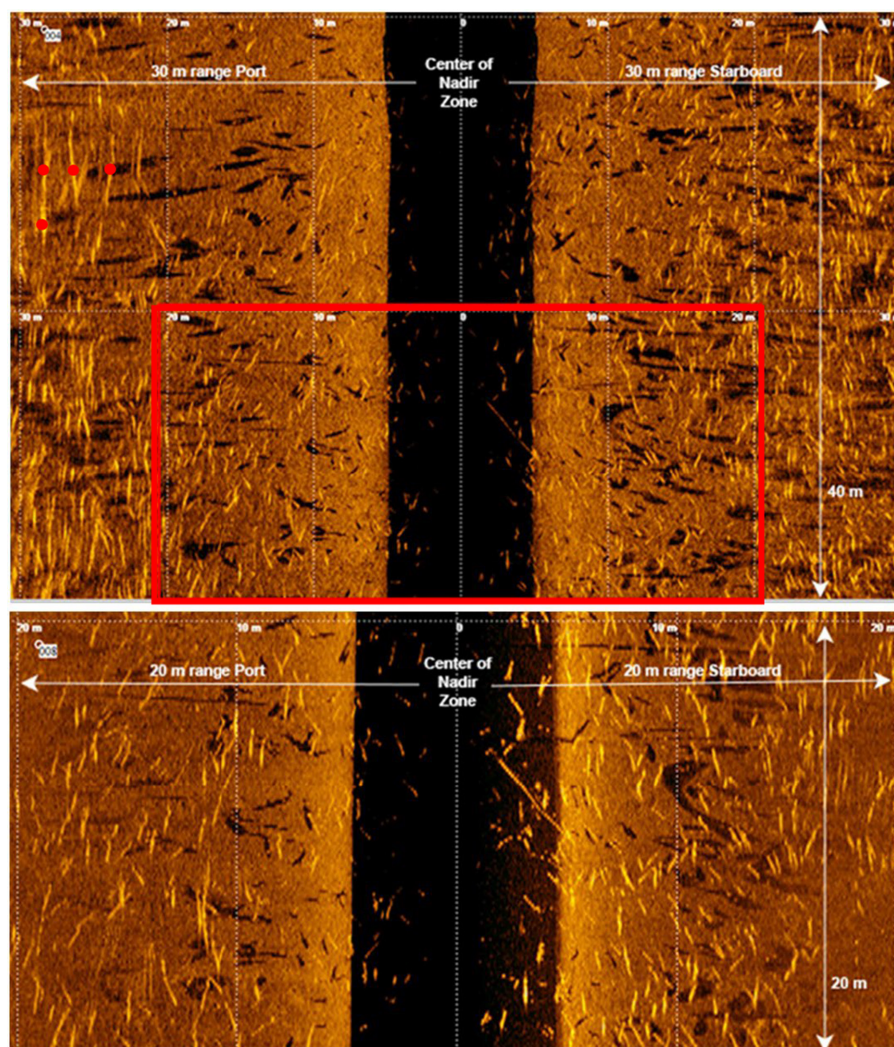


Figure 3. Example of survey-grade side-scan sonar imagery of bigheaded carp targets. The top image is from lower frequency (600-kHz) data. The white grid in the top image is 10 m across-track (to left and right) and 20 m along-track (to top and bottom). The red bounding box indicates the acoustic footprint of the higher resolution (1,600 kHz) in the lower image. The white grid in the lower image is 10 m across-track (to left and right) and 20 m along-track (to top and bottom). Note that some of the targets on the port side in the top image (denoted by red dots) appear to be swimming relative to the vessel and are elongated as a result.

data. Therefore, more time can be spent collecting survey data as opposed to processing those data. Algorithms can be retrained, refined, advanced with new technology, and improved and tested over time and across multiple data sets with changing complexity in substrates and habitats. Lawson et al. (2020), for example, used classic image processing algorithms to apply consistent scoring across a data set, which resulted in high agreement with manual counts (Spearman's rank correlation coefficient = 0.95). Andrews et al. (2020) used supervised maximum likelihood classification to manually delineate fish targets from the substrate for training a machine learning algorithm, which also resulted in high agreement relative to manual selections (kappa coefficient = 0.98). Regardless of the approach, performance metrics of automated processes are a critical component of reducing reader uncertainty. In addition, it is best practice for researchers to report the resulting performance metrics so that the validity of various automated approaches is readily available for literature review.

Statistical Analysis and Interpretation

Any number of statistical analyses may be performed on sonar data once the targets are identified (see Table 1 for published examples). The analytical approach should be determined during the design phase, although modifications are sometimes necessary to adapt to changes in the study design, the nature of the resulting data, or the availability of new information. Although a full coverage of statistical approaches is beyond the scope of this paper, we will review some common metrics and key assumptions.

Estimating abundance is usually the primary metric for sonar surveys of fishes. Counts can be made in a subset of the occupied habitat(s) and subsequently extrapolated or scaled to broader areas. Detection probability can either be estimated or assumed perfect, depending on the species, habitat, survey methodology, and data availability. The region of water that is ensonified (relative to beam angle, swath range, and transducer depth) is fundamental to detecting a particular target species. Although this is unlikely to be a major

factor for surveys of demersal fishes near the substrate, fish species (e.g., bigheaded carps) inhabiting the upper water column could be missed if the transducer is deployed too deep (e.g., when using a towfish). When considering these factors, it could be assumed that all fish present in the sonar swath are available for detection. However, establishment of an optimal range is recommended to account for signal attenuation (Figure 1) when applying extrapolation statistics.

Sizing and shape criteria for fish targets can be used to subset counts by species groups and to make more precise inferences on the target species. Fish size composition data may be applied to apportion fish targets in the imagery. However, fish orientation, swimming, and body morphology may cause size biases and warrant caution or estimation. Fish targets in the sonar imagery are comprised of pixels and are converted to a standard unit of measurement using a known calibration factor relative to the resulting resolution (e.g., pixels/mm) of the sonar imagery. Resolution is a function of several instrument variables, including signal frequency, pulse length, and beam width (i.e., primary variables of across-track resolution) as well as ping rate and survey speed (i.e., primary variables of along-track resolution). Modern SSS systems incorporate compressed high-intensity radar pulse (CHIRP) functionality in which the signal frequency is quickly ramped up from low to high (within a narrow bandwidth) to increase the pulse width. This capability increases the resolution, target separation, and effective range (Atherton 2011). For example, a 1,200-kHz CHIRP SSS equates to approximately 2 cm/pixel (Rivera et al. 2023). See Atherton (2011) for a detailed description of SSS operations and fundamentals.

We are unaware of SSS validation studies for fish size, but length measurements in other sonar types are known to vary with fish orientation (Cook et al. 2019). Targets aligned perpendicular to the sonar beam reflect the most acoustic energy for accurate measurement, but fish are rarely oriented in one direction, particularly in open, nonflowing habitats. However, orientation can be measured, and it may be plausible to apply a known correction factor if improved accuracy is needed. In addition, fish swimming along-track with the survey vessel can cause targets to be superficially elongated (e.g., Figure 3). This phenomenon is common for pelagic schooling species, such as bigheaded carps. Inducing fish to school and swim rapidly from the survey vessel can cause fish targets to be difficult to render. Depending on the species of interest, it may be possible to mitigate avoidance behavior by surveying at night and/or by using quiet survey techniques (e.g., Figure 2). We caution that some level of error is likely when measuring fish size, but it may be acceptable depending on the species of interest and the measurement precision required to meet research objectives.

Sonar is unique in its ability to sample water volume and index fish counts in several ways depending on the management or research objective. Side-scan sonar imagery is a two-dimensional transformation of a three-dimensional space, depicting an overhead view of the area surveyed. The ensonified region is dependent on the geometric relationships between transducer depth, beam angle, and across-track and along-track ranges (see manufacturer specifications). Water column depth in SSS imagery could be assumed homogeneous to depth values recorded under the survey vessel (e.g., recreational-grade systems can simultaneously collect SSS, down-imaging sonar, and depth data)

in large, flat-bottomed environments, such as those present in lakes and reservoirs (Long et al. 2024). However, bathymetry data may need to be incorporated to account for water depth variation and improve precision in the estimated volume ensonified and the corresponding fish per unit of volume (i.e., density). Bathymetry of the SSS swath may be modeled in several ways, such as interpolating along-track depth data across a series of parallel surveys (e.g., Ridgway et al. 2023), using a SSS with an interferometric system (e.g., Halmai et al. 2020), or using a separate multi-beam bathymetric sonar system. Once the ensonified volume is corrected for heterogeneous depth, known length and weight models can be applied to convert values into biomass, although this approach warrants caution when fish length measures in sonar have not been validated or verified using complementary data. Successful application of SSS for measuring fish populations is contingent on the ability to meet or otherwise address several key assumptions. Sonar is not expected to be a singular solution for addressing fisheries management needs. However, SSS is well suited for use in conjunction with any number of other survey data methods, including but not limited to fish community capture data, telemetry, and environmental DNA, which can provide an array of complementary data sources.

Opportunities for Validation

At present, there are few published examples in which SSS has gone through rigorous validation for fisheries applications (but see Fleming et al. 2018; Wolfenkoehler et al. 2023). Targets meeting certain criteria are generally assumed to represent the focal species. However, uncertainty may arise with respect to detection probability, target identity, and target size. One approach to validating SSS with respect to these metrics is to scan known targets, such as tethered fish, artificial models of fish (e.g., Wolfenkoehler et al. 2023), or other inanimate objects. These approaches may allow rigorous quantification of detection probability and measurement accuracy. However, the assumption that parameter estimates translate to real-world applications should be clearly stated or otherwise tested if possible. We note that in our experience, deceased fish may not yield satisfactory imagery when they are softer bodied and thus less reflective than a live, free-swimming animal. Research ponds containing live fish have been used to verify the size and abundance of a target species, with successful translation to a wild population (Fleming et al. 2018). Direct validation of the study species in the study system is often resource prohibitive—and, in most cases, impossible—particularly in large, dynamic systems. However, it is important to note that a variety of other data sources (e.g., traditional fish capture/sampling, acoustic telemetry, mark-recapture, optical video, and scuba surveys) has been applied successfully to confirm the estimated presence, relative abundance, or absolute abundance of the target species within the surveyed area (Bollinger and Kline 2017; Fleming et al. 2018; Mora et al. 2018; Andrews et al. 2020; Kazyak et al. 2020).

FUTURE DIRECTIONS

Future technological advancements in sonar products and data processing may provide opportunities for improving use in fisheries applications. Classic image processing algorithms have been demonstrated to be useful in automating fish detection and quantification (Lawson et al. 2020). However, AI technology provides a more robust approach

and may be warranted when dealing with large data sets containing complex environmental features and a diverse gradient of aquatic animals. Few examples of machine learning/deep learning applications for SSS currently exist (Schneider and Zhuang 2020). However, emerging applications have been applied to detect, track, and count fish targets in other sonar types (e.g., multi-beam, forward-looking sonar; Tarling et al. 2022; Wei et al. 2022). Although motion features in still imagery such as SSS are not as directly available for target recognition, repeated surveys could be used to produce training data with fine-scale abundance changes or the relocation of fish while the positioning of substrate features remains constant. Furthermore, AI advancements in other fields of study could be transferable. The medical research community has seen an exponential increase in the number of AI-based systems that are used to process extensive image data sets of biological features. For instance, more than 3,500 academic papers published (i.e., PubMed) in 2023 applied AI to process ultrasound imagery (Tenajas et al. 2023). Ultrasound technology is fundamentally the same as sonar, producing imagery with sound waves but at different frequency ranges. Over the next few years, the emergence of robust AI-based systems to process sonar data will require research collaborations across a diversity of disciplines, including computational scientists, physics experts, signal-processing engineers, and fisheries researchers.

Sonar technology is progressing at a rapid pace. Next-generation product lines and capabilities are emerging every few years, particularly in the recreational-grade sector, where equipment costs are substantially lower. Most recreational-grade SSS systems simultaneously collect down-scan sonar imagery, which has recently been applied to estimate fish density in deepwater environments (Long et al. 2024). Some recent recreational-grade models include interferometric SSS, which may be applied to create high-resolution bathymetric maps of the SSS swath (Halmai et al. 2020) and to produce three-dimensional-like SSS imagery of fish targets situated above the substrate. The recent emergence of recreational-grade live-imaging sonar (a multi-beam technology) has been embraced by the angling community, with its unique video camera-like display of fish interacting with, pursuing, and/or biting fishing lures in real time. This low-cost technology may also be applied to fish behavior research as well as deployed in conjunction with SSS surveys to provide complementary fish abundance and/or composition data. It is important to note that tangential recreational-grade sonar technologies will likewise require efficient data acquisition procedures. With greater emphasis on survey design as well as development of large, high-quality training data sets for deep learning systems, we anticipate that SSS will provide efficient approaches to address pressing questions in fisheries science.

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